End-to-End Learned Communication System: Joint Optimization of Constellation and Receiver over an AWGN Channel

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Problem Statement

Traditional communication systems rely on block-based designs with mathematically predefined modulation and demodulation schemes. This project investigates whether a deep learning model, structured as a channel autoencoder, can jointly learn its own optimal transmission constellation and receiver architecture from scratch. The core challenge is to optimize this end-to-end system over an Additive White Gaussian Noise (AWGN) channel while strictly enforcing a peak-power constraint, and to evaluate if the learned representations can match or outperform classical digital modulation schemes.

System to be Simulated

The system models a complete physical layer communication link as a neural network autoencoder. The transmitter (encoder) maps discrete input messages into continuous modulated symbols. A normalization layer is applied to enforce a strict peak-power constraint before transmission. These symbols then traverse a simulated AWGN channel layer, which adds stochastic noise to the signal. Finally, the receiver (decoder network) observes the noisy channel outputs and computes a probability distribution to recover the original transmitted messages.

Methodology

The autoencoder will be implemented entirely in Python utilizing deep learning frameworks (e.g., PyTorch or TensorFlow). Both the transmitter and receiver will be constructed using fully connected neural network layers. The end-to-end model will be trained jointly using stochastic gradient descent and categorical cross-entropy loss to minimize the message reconstruction error. Following the training phase, the intermediate layer activations (representing the learned constellation points) will be extracted to visualize the physical layer representation learned by the network.

Expected Results

We expect the autoencoder to successfully converge and learn a highly customized signal space. Under the peak-power constraint, the visualization of the learned constellations is anticipated to be visually striking and non-standard, adapting intuitively to the channel and power limitations. The headline result will be trade-off curves comparing Symbol Error Rate (SER) versus Signal-to-Noise Ratio (SNR), demonstrating the learned system's efficiency.

Validation Against Theory

The simulation will be rigorously validated by benchmarking the autoencoder's SER performance directly against the theoretical and simulated error rates of classical modulation schemes, specifically QPSK and 16-QAM. This comparison will be conducted at an equal transmission rate to quantify the exact performance delta between machine-learned and human-engineered constellations.